HW 6 — Jackson Eq. 8.59 (TE) & Rectangular Cavity

black = write it down · **red = the answer** · *teal = my note, do not copy* · grey = optional

Q1 Derive Jackson **Eq. 8.59** for **TE** modes — Lec 13 s2

Q2 Rectangular resonant cavity — general **TM** solution and lowest resonant frequency — Lec 14 s14–15

Both follow lectures you already have. Q1: Lec 13 slide 2 prints Eq. 8.59 and then uses it — this question asks for the derivation behind the TE line. Q2: Lec 14 slide 15 works the **cylindrical** cavity with Bessel functions; this is the **rectangular** twin, so sines replace the Bessel functions and everything else is identical.

Checked numerically. Q1: the identity $|\hat{n}\times\mathbf{H}|^2 = (k^2/\gamma^4)|\hat{n}\times\nabla_t\psi|^2 + |\psi|^2$ was tested on all four walls of a TE_{10} guide (exact), and the algebra conversion at three frequencies (exact). Q2: the cavity fields satisfy both Maxwell curl equations and $\nabla\cdot\mathbf{E} = 0$, and all six wall conditions hold to 10^{-16}.

Question 1 — Jackson Eq. 8.59 for TE modes

> Show that for a **TE** mode the power lost per unit length of guide is $-dP/dz = (1/2\sigma\delta)(\omega/\omega_\lambda)^2 \oint_C [\,(1/\mu\varepsilon\omega_\lambda^2)(1 - \omega_\lambda^2/\omega^2)\,|\mathbf{n}\times\nabla_t\psi|^2 + (\omega_\lambda^2/\omega^2)|\psi|^2\,]\,dl$.

ANSWER

$$-\frac{dP}{dz} = \frac{1}{2\sigma\delta}\left(\frac{\omega}{\omega_\lambda}\right)^2 \oint_C \left[\frac{1}{\mu\varepsilon\omega_\lambda^2}\left(1 - \frac{\omega_\lambda^2}{\omega^2}\right)\left|\hat{n}\times\nabla_t\psi\right|^2 + \frac{\omega_\lambda^2}{\omega^2}\left|\psi\right|^2\right] dl$$

Flow: start from the wall-loss formula → work out $\hat{n}\times\mathbf{H}$ on the wall, which splits into **two** perpendicular pieces → square it → rewrite the prefactor in terms of ω/ω_λ.

Step 0 — The starting formula

$$-\frac{dP}{dz} = \frac{1}{2\sigma\delta}\oint_C \left|\hat{n}\times\vec{H}\right|^2 dl$$ Jackson 8.58

So the whole job is to compute $\left|\hat{n}\times\vec{H}\right|^2$ on the wall.

Step 1 — The TE fields

For TE, $E_z = 0$ and the mode is built from $\psi = H_z$:

$$\vec{H} = \vec{H}_t + \hat{z}\psi, \qquad \vec{H}_t = \frac{ik}{\gamma^2}\nabla_t\psi$$ Lec 10, s18

wall condition: $\dfrac{\partial\psi}{\partial n} = 0$ on C *this is used in Step 3*

*Unlike TM, here **$H_z \neq 0$ at the wall** — that is exactly why the TE answer has an extra $|\psi|^2$ term that TM does not.*

Step 2 — Work out $\hat{n}\times\vec{H}$ at the wall

Set up a right-handed triad on the wall: $\hat{n}$ (outward normal, lies in the cross-section), $\hat{t}$ (along the wall), and $\hat{z}$:

$$\hat{n}\times\hat{t} = \hat{z}, \quad \hat{t}\times\hat{z} = \hat{n}, \quad \hat{z}\times\hat{n} = \hat{t} \quad \Rightarrow \quad \hat{n}\times\hat{z} = -\hat{t}$$

Now cross $\hat{n}$ into the two pieces of $\mathbf{H}$ separately:

$$\hat{n}\times\vec{H} = \hat{n}\times\vec{H}_t + \psi(\hat{n}\times\hat{z})$$

Write $\mathbf{H}_t$ in the wall frame as $(\mathbf{H}_t\cdot\hat{n})\hat{n} + (\mathbf{H}_t\cdot\hat{t})\hat{t}$. The $\hat{n}$ part dies under $\hat{n}\times$, leaving

$$\hat{n}\times\vec{H} = (\vec{H}_t\cdot\hat{t})\hat{z} - \psi\hat{t}$$

*The two surviving pieces point along $\hat{z}$ and $\hat{t}$ — **perpendicular to each other**. So when we square, there is no cross term; the magnitudes just add in quadrature.*

Step 3 — Square it

$$\left|\hat{n}\times\vec{H}\right|^2 = \left|\vec{H}_t\cdot\hat{t}\right|^2 + \left|\psi\right|^2$$

For the first piece, put in $\mathbf{H}_t$ from Step 1:

$$\vec{H}_t\cdot\hat{t} = \frac{ik}{\gamma^2}(\nabla_t\psi\cdot\hat{t}) = \frac{ik}{\gamma^2}\frac{\partial\psi}{\partial t}$$

Here is where the TE wall condition enters. Since $\partial\psi/\partial n = 0$ on C, the gradient at the wall has **no normal part** — it points purely along $\hat{t}$:

$$\nabla_t\psi = \frac{\partial\psi}{\partial t}\hat{t} \quad \Rightarrow \quad \hat{n}\times\nabla_t\psi = \frac{\partial\psi}{\partial t}\hat{z} \quad \Rightarrow \quad \left|\frac{\partial\psi}{\partial t}\right| = \left|\hat{n}\times\nabla_t\psi\right|$$

$$\Rightarrow \quad \left|\hat{n}\times\vec{H}\right|^2 = \frac{k^2}{\gamma^4}\left|\hat{n}\times\nabla_t\psi\right|^2 + \left|\psi\right|^2$$

Tested numerically on all four walls of a TE_{10} rectangular guide — exact every time.

Step 4 — Put the prefactor into Jackson's form

We now have

$$-\frac{dP}{dz} = \frac{1}{2\sigma\delta} \oint_C \left[\frac{k^2}{\gamma^4} \left|\hat{n}\times\nabla_t\psi\right|^2 + \left|\psi\right|^2 \right] dl$$

Jackson pulls out a factor $(\omega/\omega_\lambda)^2$, so we need to rewrite each term with that factored out.

Use the two standard relations:

$$k^2 = \mu\varepsilon\omega^2 - \gamma^2, \qquad \gamma^2 = \mu\varepsilon\omega_\lambda^2$$ Lec 11, s2–4

First term. Write $k^2 = \mu\varepsilon\omega^2(1 - \omega_\lambda^2/\omega^2)$ and $\gamma^4 = (\mu\varepsilon)^2\omega_\lambda^4$:

$$\frac{k^2}{\gamma^4} = \frac{\mu\varepsilon\omega^2(1 - \omega_\lambda^2/\omega^2)}{(\mu\varepsilon)^2\omega_\lambda^4}$$

Split off $\omega^2/\omega_\lambda^2$ and cancel one $\mu\varepsilon$:

$$= \left(\frac{\omega}{\omega_\lambda}\right)^2 \frac{1}{\mu\varepsilon\omega_\lambda^2} \left(1 - \frac{\omega_\lambda^2}{\omega^2}\right)$$ ✓ Jackson's first TE term

Second term. The $\left|\psi\right|^2$ has coefficient 1, and

$$1 = \left(\frac{\omega}{\omega_\lambda}\right)^2 \frac{\omega_\lambda^2}{\omega^2}$$ ✓ Jackson's second TE term — just 1 written the useful way

Both terms now carry the same $(\omega/\omega_\lambda)^2$, so it comes outside the integral:

ANSWER — Q1

$$-\frac{dP}{dz} = \frac{1}{2\sigma\delta} \left(\frac{\omega}{\omega_\lambda}\right)^2 \oint_C \left[\frac{1}{\mu\varepsilon\omega_\lambda^2} \left(1 - \frac{\omega_\lambda^2}{\omega^2}\right) \left|\hat{n}\times\nabla_t\psi\right|^2 + \frac{\omega_\lambda^2}{\omega^2} \left|\psi\right|^2 \right] dl$$ ✓

OPTIONAL **Why TM looks different.** For TM the wall condition is $\psi = 0$, so the $\left|\psi\right|^2$ term vanishes — but $\partial\psi/\partial n \neq 0$, so the surviving piece is $\left|\partial\psi/\partial n\right|^2$. That is Jackson's upper line.

■ END OF QUESTION 1 ■

Question 2 — Rectangular resonant cavity, TM modes

> A resonant cavity of perfect conductor has a **rectangular** cross section with sides a and b (and length d along z). Find the general solution for **TM** modes, and the **lowest** resonant frequency.

ANSWER

$$E_z = E_0 \sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\cos\left(\frac{p\pi z}{d}\right), \qquad H_z = 0$$

$$\omega_{mnp} = \frac{1}{\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2} \qquad m, n = 1, 2, 3\ldots \quad p = 0, 1, 2\ldots$$

$$\textbf{lowest} = TM_{110}: \quad \omega_{110} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$$

Flow: a cavity is a waveguide with two end caps bolted on → so solve the waveguide first (that gives the x, y shape) → then the wave cannot escape, it bounces, so put **both** directions ±z in and make the end caps happy (that gives the z shape) → now all three directions are quantised, and that fixes ω.

Step 0 — What we start from

Source-free region inside, time dependence $e^{-i\omega t}$:

$$\nabla\times\vec{E} = i\omega\mu\vec{H}, \qquad \nabla\times\vec{H} = -i\omega\varepsilon\vec{E}, \qquad \nabla\cdot\vec{E} = 0, \qquad \nabla\cdot\vec{H} = 0$$ Lec 9, s11

The walls are perfect conductor, so on **every** wall (all six faces)

$$\hat{n}\times\vec{E} = 0 \quad \textit{tangential E vanishes} \qquad \hat{n}\cdot\vec{H} = 0 \quad \textit{normal H vanishes}$$

Why these two. *Inside a perfect conductor E = 0 and H = 0. Tangential E and normal H are the two combinations that must be continuous across a surface, so they are zero on our side as well. Normal E and tangential H are not continuous — they are paid for by surface charge and surface current, so they are free to be non-zero.*

And TM means, by definition,

$$H_z = 0, \qquad \psi \equiv E_z \quad \textit{the one scalar everything is built from}$$

Step 1 — The waveguide piece: shape in x and y

Take the curl of the first Maxwell equation and use the second plus $\nabla\cdot\mathbf{E} = 0$; every component then obeys the wave equation. Put $E_z = \psi(x,y)\,Z(z)$:

$$(\nabla^2 + \mu\varepsilon\omega^2)E_z = 0 \quad \Rightarrow \quad \nabla_t^2\psi + \gamma^2\psi = 0, \qquad \gamma^2 = \mu\varepsilon\omega^2 - k^2$$ Lec 10, s17

with $\nabla_t^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2$, and k the z wavenumber. E_z is **tangential** to the four side walls, so

$$\psi = 0 \quad \text{at} \quad x = 0, a \quad \text{and} \quad y = 0, b$$

Separate $\psi = X(x)Y(y)$. Dividing by XY gives $X''/X + Y''/Y = -\gamma^2$; one term depends only on x, the other only on y, so each must be its own constant:

$$X'' = -k_x^2 X, \qquad Y'' = -k_y^2 Y, \qquad k_x^2 + k_y^2 = \gamma^2$$

X is a sine or a cosine. $X(0) = 0$ rules out the cosine, so $X = \sin(k_x x)$; then $X(a) = 0$ forces $k_x a = m\pi$. Identically for Y:

$$\psi = E_0 \sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right), \qquad \gamma^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

*$m = 0$ or $n = 0$ kills ψ completely ($\sin 0 = 0$), so **$m, n \geq 1$** already at this step.*

Step 2 — The other components, in terms of ψ

For one travelling wave, $Z = e^{ikz}$, Maxwell gives the TM relations

$$\vec{E}_t = \frac{ik}{\gamma^2}\,\nabla_t\psi, \qquad \vec{H}_t = \frac{i\omega\varepsilon}{\gamma^2}\,(\hat{z}\times\nabla_t\psi)$$ Lec 10, s17

In a cavity the wave bounces off both end caps, so we need **both** directions:

$$Z(z) = A\,e^{ikz} + B\,e^{-ikz}$$

Add the two travelling waves. $\mathbf{E}_t$ carries a factor ik, which flips sign for the $-k$ wave, and that signed ik times $e^{\pm ikz}$ is exactly dZ/dz. $\mathbf{H}_t$ carries no k at all, so it simply adds:

$$\vec{E}_t = \frac{1}{\gamma^2}\,\nabla_t\psi\,\frac{dZ}{dz}, \qquad \vec{H}_t = \frac{i\omega\varepsilon}{\gamma^2}\,(\hat{z}\times\nabla_t\psi)\,Z$$

Line by line: $Z' = ikA\,e^{ikz} - ikB\,e^{-ikz}$, which is $(ik)(A\,e^{ikz}) + (-ik)(B\,e^{-ikz})$ — the $+ik$ of the forward wave and the $-ik$ of the backward one. So writing dZ/dz handles both at once.

Step 3 — The end caps: shape in z

On the cap at $z = 0$ the normal is $\hat{z}$, so the tangential field there is $\mathbf{E}_t$ (that is E_x and E_y). Same at $z = d$. So

$$\vec{E}_t = 0 \quad \text{at} \quad z = 0, d \quad \Rightarrow \quad \frac{dZ}{dz} = 0 \quad \text{at} \quad z = 0, d$$ *Step 2, since $\nabla_t \psi \neq 0$*

E_z is **normal** to the caps, so it is not forced to vanish there — which is why z comes out as a cosine and not a sine. Rewrite $Z = A\,e^{ikz} + B\,e^{-ikz}$ as the equivalent pair $Z = C \cos kz + D \sin kz$, and differentiate:

$$Z' = -Ck \sin kz + Dk \cos kz$$

condition	gives	result
$Z'(0) = 0$	$Dk = 0$	$D = 0$, so $Z = C \cos kz$
$Z'(d) = 0$	$-Ck \sin kd = 0$	$kd = p\pi$, $p = 0, 1, 2\ldots$

$$\Rightarrow \quad Z(z) = \cos\left(\frac{p\pi z}{d}\right), \qquad k = \frac{p\pi}{d}$$ *C absorbed into E_0*

p = 0 is allowed *here (cos 0 = 1 survives), unlike m and n. At p = 0 we get Z′ = 0, so $\mathbf{E}_t$ = 0: only E_z and $\mathbf{H}_t$ remain, both independent of z.*

Step 4 — The resonant frequency

Step 1 defined $\gamma^2 = \mu\varepsilon\omega^2 - k^2$. Solve it for ω^2 and substitute both quantised results:

$$\omega^2 = \frac{1}{\mu\varepsilon}\left[\gamma^2 + k^2\right] = \frac{1}{\mu\varepsilon}\left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2\right]$$ same form as Lec 14, s14

This is the whole difference between a guide and a cavity. In a guide you pick ω and k adjusts to it. Here the caps pin k to pπ/d, and γ is already pinned by a and b, so nothing is left free — ω can only take the discrete values below. That is what "resonant" means.

Step 5 — Write out all six components

Put ψ from Step 1 and Z from Step 3 into Step 2, using $\nabla_t\psi = (\partial\psi/\partial x, \partial\psi/\partial y, 0)$ and $\hat{z}\times\nabla_t\psi = (-\partial\psi/\partial y, \partial\psi/\partial x, 0)$:

ANSWER — general TM solution

$$E_z = E_0 \sin(m\pi x/a)\sin(n\pi y/b)\cos(p\pi z/d)$$

$$E_x = -(p\pi/d)(m\pi/a)(E_0/\gamma^2)\cos(m\pi x/a)\sin(n\pi y/b)\sin(p\pi z/d)$$

$$E_y = -(p\pi/d)(n\pi/b)(E_0/\gamma^2)\sin(m\pi x/a)\cos(n\pi y/b)\sin(p\pi z/d)$$

$$H_x = -i\omega\varepsilon(n\pi/b)(E_0/\gamma^2)\sin(m\pi x/a)\cos(n\pi y/b)\cos(p\pi z/d)$$

$$H_y = +i\omega\varepsilon(m\pi/a)(E_0/\gamma^2)\cos(m\pi x/a)\sin(n\pi y/b)\cos(p\pi z/d)$$

$$H_z = 0$$

$\gamma^2 = (m\pi/a)^2 + (n\pi/b)^2$, $m, n = 1, 2, 3\ldots$, $p = 0, 1, 2\ldots$, everything $\times e^{-i\omega t}$.

Check the six faces. On each face, only the two components lying in that face have to vanish:

face	tangential E there	why it is zero
$x = 0, a$	E_y, E_z	both carry $\sin(m\pi x/a)$
$y = 0, b$	E_x, E_z	both carry $\sin(n\pi y/b)$
$z = 0, d$	E_x, E_y	both carry $\sin(p\pi z/d)$

And $\hat{n}\cdot\mathbf{H} = 0$ comes for free: $H_z = 0$ takes care of the two caps, H_x carries $\sin(m\pi x/a)$, and H_y carries $\sin(n\pi y/b)$.

Step 6 — The lowest resonant frequency

ω_{mnp} increases with every index, so take each index as small as it is allowed to be:

index	smallest allowed	reason
m	1	m = 0 makes $\sin(m\pi x/a) \equiv 0$ — no field at all
n	1	same, $\sin(n\pi y/b) \equiv 0$
p	0	$\cos 0 = 1$, so E_z and H_t survive

So the lowest mode is TM_{110}. Put m = n = 1, p = 0 into Step 4:

ANSWER — lowest resonant frequency

$$\omega_{110} = \frac{1}{\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$$

Independent of the cavity length d, because p = 0.

Same pattern as the lecture. Lec 14 s15 does the **cylindrical** cavity and gets $\omega_{TM010} = 2.405/(R\sqrt{\mu\varepsilon})$ — also p = 0, also set by the cross-section alone. There the x, y shape was a Bessel function with first zero $x_{01} = 2.405$; here it is a pair of sines giving π/a and π/b. Every other step is the same.

OPTIONAL **Numbers, as a check.** For a = 2.3 cm, b = 1.1 cm, d = 3.4 cm, $\mu = 1.4\mu_0$, $\varepsilon = 2.6\varepsilon_0$, the ordering comes out TM_{110} (8.11 GHz) < TM_{111} (8.72) < TM_{112} (10.2) < TM_{210} (10.6) — TM_{110} is lowest, as claimed.

■ END OF REQUIRED SOLUTION ■